

The block diagram illustrates the hardware components and their connections:

- Battery 5v** is connected to the **USB-C UART** module.
- The **USB-C UART** module is connected to the **Waveshare LCD (800x480) ESP32-S3** module.
- The **Waveshare LCD (800x480) ESP32-S3** module is connected to the **ADC** and **I2C** modules.
- The **ADC** module is connected to the **IO 6** pin.
- The **I2C** module is connected to the **IO 6** pin.

